

# Visualization Literacy Analysis

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## Read in data files

Raw data is in the ../AccData subdirectory in two excel spreadsheet files per participant. The following code combines into single dataframe of all data.

```
#grab files from google drive (only have to do this once)
# source("getFromGoogleDrive.R")

#get the first 6? characters of each data file
#get unique values of these
# this is list of subject ids

raw_file_names <- list.files("../AccData")
first_six <- substr(raw_file_names, 1, 6)
sub_ids <- unique(first_six)

# fast_RT = data.frame(ParticipantId = character(),
#                        TrialName = character(),
#                        type = character(),
#                        time = numeric()
# )

all_data <- NULL

for (i in 1:length(sub_ids)){

  if (grepl("~$", sub_ids[i], fixed = T)){
    next
  }

  temp_file1 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_1.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  temp_file2 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_2.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  new_temp <- temp_file1 %>%
    bind_cols(temp_file2$correct) %>%
```

```

  rename(Correct_1 = correct, Correct_2 = "...9") %>%
  mutate(AnswerRT = TimeToBeginInput - TimeToReadQuestion)

all_data <- all_data %>%
  bind_rows(new_temp)
}

all_data <- all_data %>%
  rename(readRT = TimeToReadQuestion, totalRT = TimeToBeginInput)

#read in trialtype key (I created this from an early version of the previous paper)
trial_type_key <- read.csv("../trial_type_key.csv", stringsAsFactors = F)

all_data <- all_data %>%
  mutate(TrialType = trial_type_key$TrialType[match(TrialName, trial_type_key$TrialName)]) %>%
  mutate(TrialType = paste0("Type", TrialType))

# remove temporary dataframes to keep workspace cleaner as we go forward
rm(new_temp)
rm(temp_file1)
rm(temp_file2)
rm(trial_type_key)

```

## Basic checks

Just a sanity check of all the data. Summary of number of subjects in each of the 3 condition types.

```

#how many participants per condition
all_data %>%
  group_by(ParticipantId, Condition) %>%
  summarize(ntrials = n()) %>%
  group_by(Condition) %>%
  summarize(nsubs = n()) %>%
  kable()

```

## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups` argument.

| Condition         | nsubs |
|-------------------|-------|
| VR                | 50    |
| VR Monitor        | 39    |
| VR Monitor Stereo | 33    |

```

#why is the balance so off?

```

## Remove outliers based on RT

Removing outliers, resulting in data frame named `all_data_no_outliers`.

1. All trials where the `AnswerRT` is less than 2 seconds are removed first, which assumes these reactions were too fast so were done without an actual considered response.

2. All trials where the **AnswerRT** is 3 standard deviations different from the mean are removed next.

Table shows summary of mean and standard deviation of answer RT by each of the trial question types, as well as the 3 standard deviation upper and lower bound of the reaction time.

Displayed histogram is a sanity check of the number of trials each participant in all of the data have done. Most participants should do 16 trials. Less than that are from dropped data for outliers.

```
#removing on trial-by-trial basis
```

```
#remove answerRTs below 2000ms first
```

```
all_data_remove <- all_data %>%
```

```
  filter(AnswerRT >= 2000)
```

```
dim(all_data)[1]
```

```
## [1] 2074
```

```
dim(all_data_remove)[1]
```

```
## [1] 2067
```

```
#drops 7 trials
```

```
# also remove AnswerRT that are more than 3 standard deviations away from
```

```
# the mean
```

```
all_data_no_outliers <- all_data_remove %>%
```

```
  group_by(TrialName) %>%
```

```
  filter(!(abs(AnswerRT - mean(AnswerRT)) > 3.0*sd(AnswerRT)))
```

```
dim(all_data_no_outliers)[1]
```

```
## [1] 2020
```

```
#drops 47 more trials
```

```
rt_data_summary <- all_data_no_outliers %>%
```

```
  group_by(TrialName) %>%
```

```
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
```

```
            sdAnswerRT = sd(AnswerRT, na.rm = T),
```

```
            LB = meanAnswerRT - 3*sdAnswerRT,
```

```
            UB = meanAnswerRT + 3*sdAnswerRT)
```

```
kable(rt_data_summary)
```

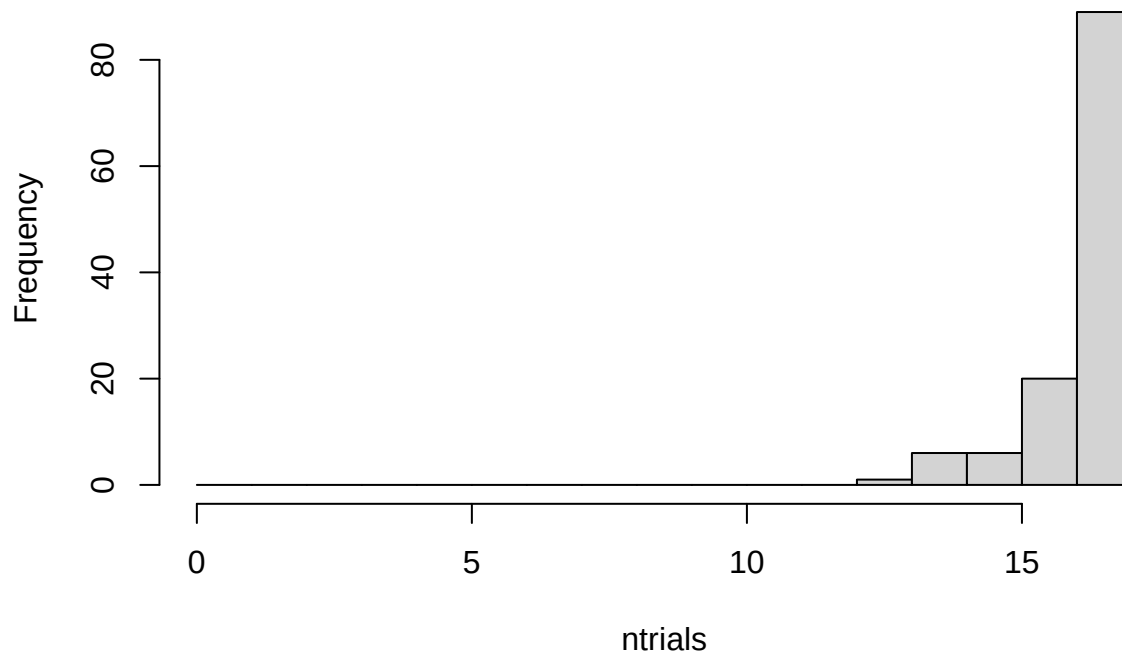
| TrialName     | meanAnswerRT | sdAnswerRT | LB         | UB        |
|---------------|--------------|------------|------------|-----------|
| BarChartQ1    | 26701.293    | 13004.715  | -12312.853 | 65715.44  |
| BarChartQ2    | 15688.149    | 10925.288  | -17087.717 | 48464.01  |
| BarChartQ3    | 14701.729    | 8225.461   | -9974.653  | 39378.11  |
| BarChartQ4    | 8962.825     | 5155.541   | -6503.798  | 24429.45  |
| LineChartQ1   | 46326.506    | 23587.872  | -24437.109 | 117090.12 |
| LineChartQ2   | 38246.295    | 23995.233  | -33739.405 | 110232.00 |
| LineChartQ3   | 25274.621    | 14260.815  | -17507.824 | 68057.07  |
| LineChartQ4   | 13372.279    | 13262.905  | -26416.437 | 53161.00  |
| LineChartQ5   | 27265.489    | 15938.188  | -20549.075 | 75080.05  |
| ScatterplotQ1 | 29913.045    | 18641.608  | -26011.778 | 85837.87  |
| ScatterplotQ2 | 27968.794    | 20282.704  | -32879.316 | 88816.90  |
| ScatterplotQ3 | 42887.765    | 26920.371  | -37873.347 | 123648.88 |

| TrialName     | meanAnswerRT | sdAnswerRT | LB         | UB        |
|---------------|--------------|------------|------------|-----------|
| ScatterplotQ4 | 33751.255    | 19808.692  | -25674.820 | 93177.33  |
| ScatterplotQ5 | 58482.379    | 38675.866  | -57545.219 | 174509.98 |
| SurfacePlotQ1 | 53283.033    | 29477.074  | -35148.188 | 141714.25 |
| SurfacePlotQ2 | 61771.846    | 43154.455  | -67691.520 | 191235.21 |
| SurfacePlotQ3 | 44518.559    | 29753.055  | -44740.607 | 133777.72 |

```
rm(rt_data_summary)

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:17))
```

**Histogram of ntrials**



```
rm(all_data_remove)
##maybe consider replacing outliers with means instead of removing them?
```

## Extract Only One Type of Chart

Create a dataframe from the data with no outliers that contains only XChartQ1 through XChartQ4 trials for the following analysis.

We replace the `all_data_no_outliers` dataframe with one containing only the selected TrialName question type here, so past this point we are analyzing only this particular TrialName type.

We show the table again of the mean RT for this question type. The table should match the one done with all of the data, but we should only have the particular question trial name type now in the data.

We also again plot the histogram of the number of trials. Participants should have seen 4 of each particular question type, but after removing outliers some only have 3, 2 or 1 remaining.

```
all_data_no_outliers <- all_data_no_outliers %>%
  filter(grepl("BarChartQ[0-9]", TrialName))

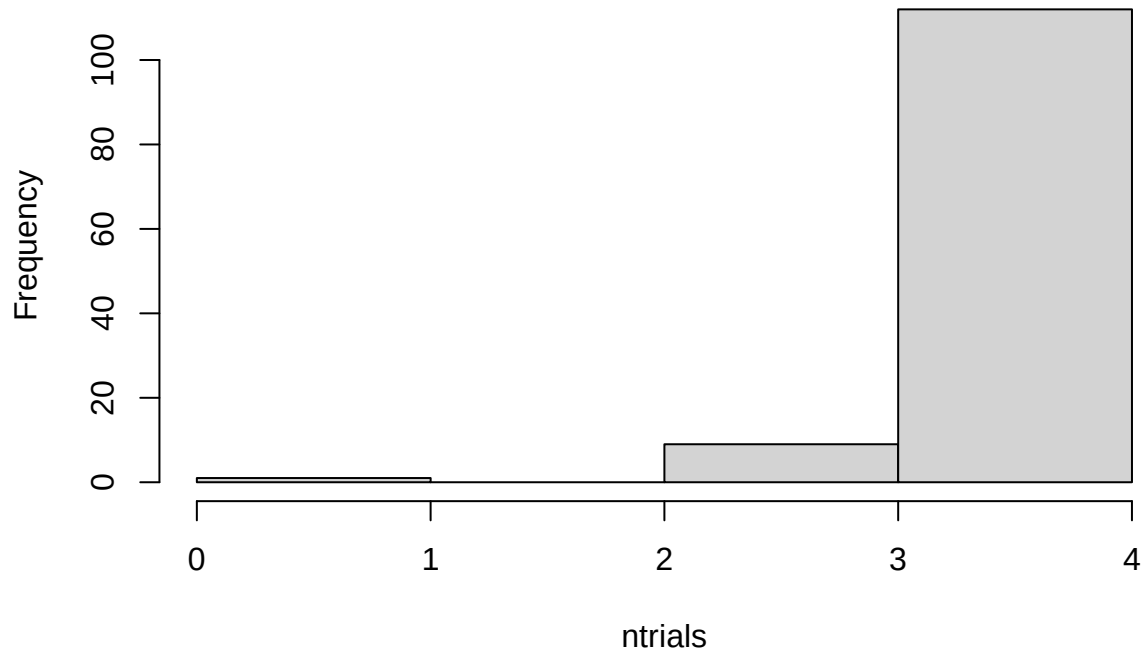
rt_data_summary <- all_data_no_outliers %>%
  group_by(TrialName) %>%
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
            sdAnswerRT = sd(AnswerRT, na.rm = T),
            LB = meanAnswerRT - 3*sdAnswerRT,
            UB = meanAnswerRT + 3*sdAnswerRT)
kable(rt_data_summary)
```

| TrialName  | meanAnswerRT | sdAnswerRT | LB         | UB       |
|------------|--------------|------------|------------|----------|
| BarChartQ1 | 26701.293    | 13004.715  | -12312.853 | 65715.44 |
| BarChartQ2 | 15688.149    | 10925.288  | -17087.717 | 48464.01 |
| BarChartQ3 | 14701.729    | 8225.461   | -9974.653  | 39378.11 |
| BarChartQ4 | 8962.825     | 5155.541   | -6503.798  | 24429.45 |

```
rm(rt_data_summary)

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:4))
```

## Histogram of ntrials



## Compare Read RT

Compare the reading reaction times for the selected question type.

### Extract wide dataframe

In this cell we convert the data to a wide format, selection only Condition, TrialType and the mean\_readRT, and widening the TrialType column.

We have also filled in na/missing values here as a final step here. Maybe we should instead just change from removing outliers to replacing with mean. Also the filling missing values code here should be easier to do, needs to be fixed. But result should be that any na in the resulting wide table is replaced with the mean for the particular Type.

```
#compare read times (should be no differences of condition)
```

```
trial_type_means <- all_data_no_outliers %>%  
  group_by(ParticipantId, Condition, TrialType) %>%  
  summarize(mean_readRT = mean(readRT), n = n())
```

```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using  
## the `.groups` argument.
```

```
# first, make .csv files in wide format to double check in statview
```

```
read_rt <- trial_type_means %>%  
  select(ParticipantId, Condition, TrialType, mean_readRT) %>%  
  pivot_wider(names_from = TrialType, values_from = mean_readRT)
```

```
# fill in na values with mean for type1, type2 and type3 RT
# the following is supposed to work, but we get NaN from calculation of mean here, not sure why...
#read_rt_type_wider_clean <- read_rt_type_wider %>% mutate(across(Type1, ~replace_na(., mean(., na.rm=T)))
read_rt <- read_rt %>% replace_na(list(Type1 = mean(read_rt$Type1, na.rm=TRUE)))
read_rt <- read_rt %>% replace_na(list(Type2 = mean(read_rt$Type2, na.rm=TRUE)))
read_rt <- read_rt %>% replace_na(list(Type3 = mean(read_rt$Type3, na.rm=TRUE)))
```

```
read_rt_summary <- read_rt %>%
  group_by(Condition) %>%
  summarize(meanType1 = mean(Type1, na.rm = T),
            sdType1 = sd(Type1, na.rm = T),
            meanType2 = mean(Type2, na.rm = T),
            sdType2 = sd(Type2, na.rm = T),
            meanType3 = mean(Type3, na.rm=T),
            sdType3 = sd(Type3, na.rm = T))
kable(read_rt_summary)
```

| Condition         | meanType1 | sdType1  | meanType2 | sdType2  | meanType3 | sdType3  |
|-------------------|-----------|----------|-----------|----------|-----------|----------|
| VR                | 6311.628  | 4683.468 | 5813.306  | 2313.923 | 4991.603  | 1971.907 |
| VR Monitor        | 6771.295  | 4183.453 | 5877.805  | 2292.265 | 5608.475  | 3055.259 |
| VR Monitor Stereo | 5642.789  | 2597.389 | 5807.304  | 1897.514 | 5267.293  | 1523.964 |

```
rm(read_rt_summary)

#rm(trial_type_means)
```

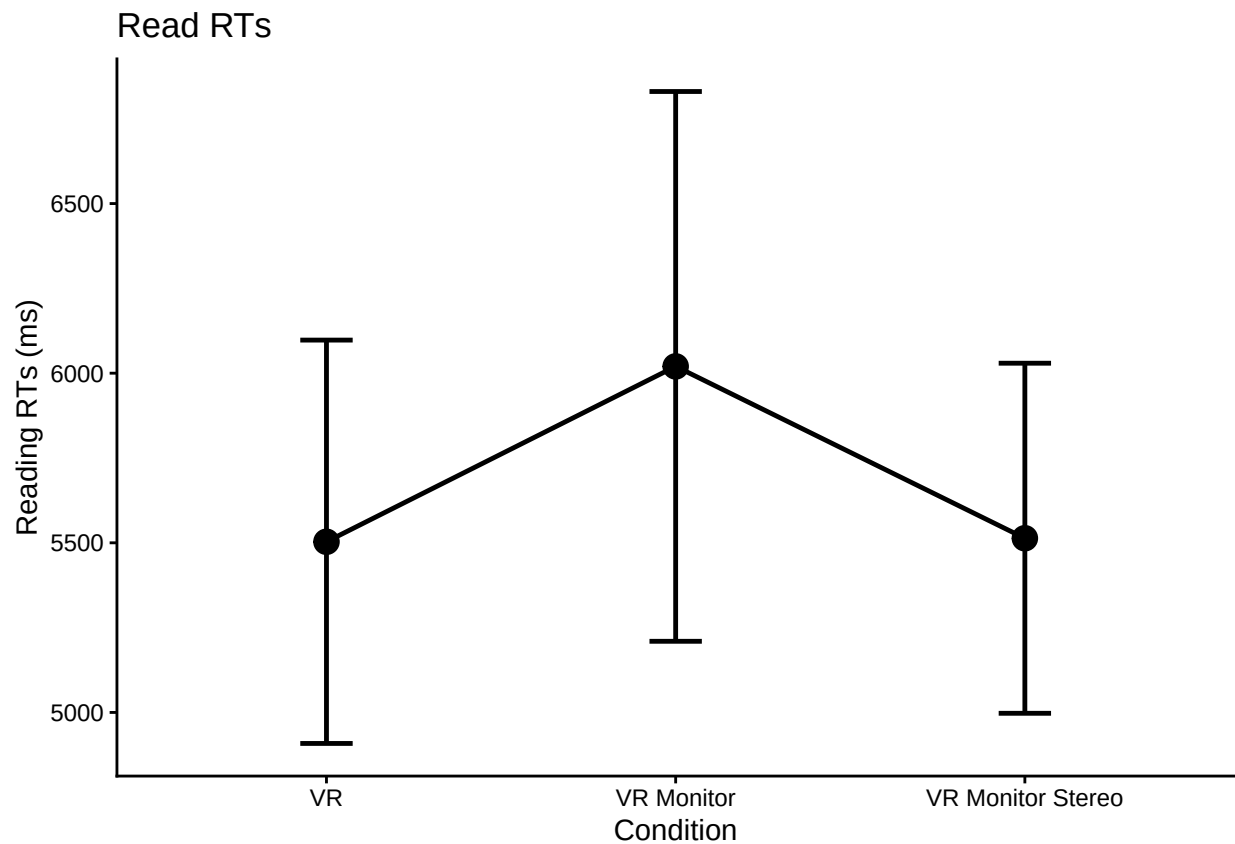
## Basic plots of main effect and wider format

Plot of read RT condition main effect, as well as wider format plot by the trial types.

```
#just condition main effect
readplot1 <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition) %>%
  summarize(overallmean = mean(readRT)) %>%
  group_by(Condition) %>%
  summarize(overall_condition_mean = mean(overallmean),
            se = std.error(overallmean),
            n = n(),
            CI = qt(0.975,df=n-1)*se)
```

## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups` argument.

```
ggplot(readplot1, aes(Condition,
                      overall_condition_mean,
                      group = 1,
                      ymin = overall_condition_mean - CI,
                      ymax = overall_condition_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Reading RTs (ms)", title = "Read RTs")
```

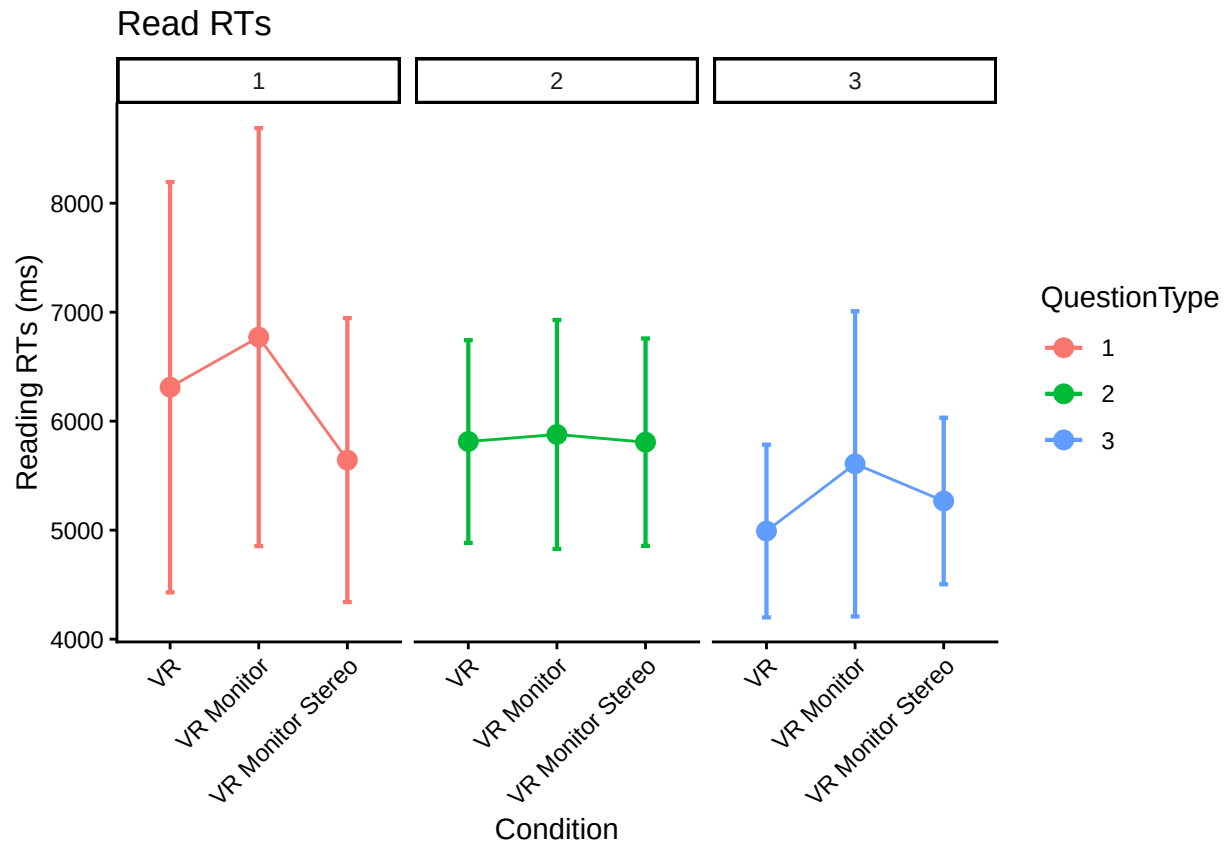


```
rm(readplot1)

#make a little plot using wide format
superbPlot(read_rt,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
  gamma = 0.95,
  adjustments = list(
    purpose = "difference"
  ),
  plotLayout = "line",
  factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1)) +
facet_wrap(vars(QuestionType)) +
labs(y = "Reading RTs (ms)", title = "Read RTs")
```

## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".



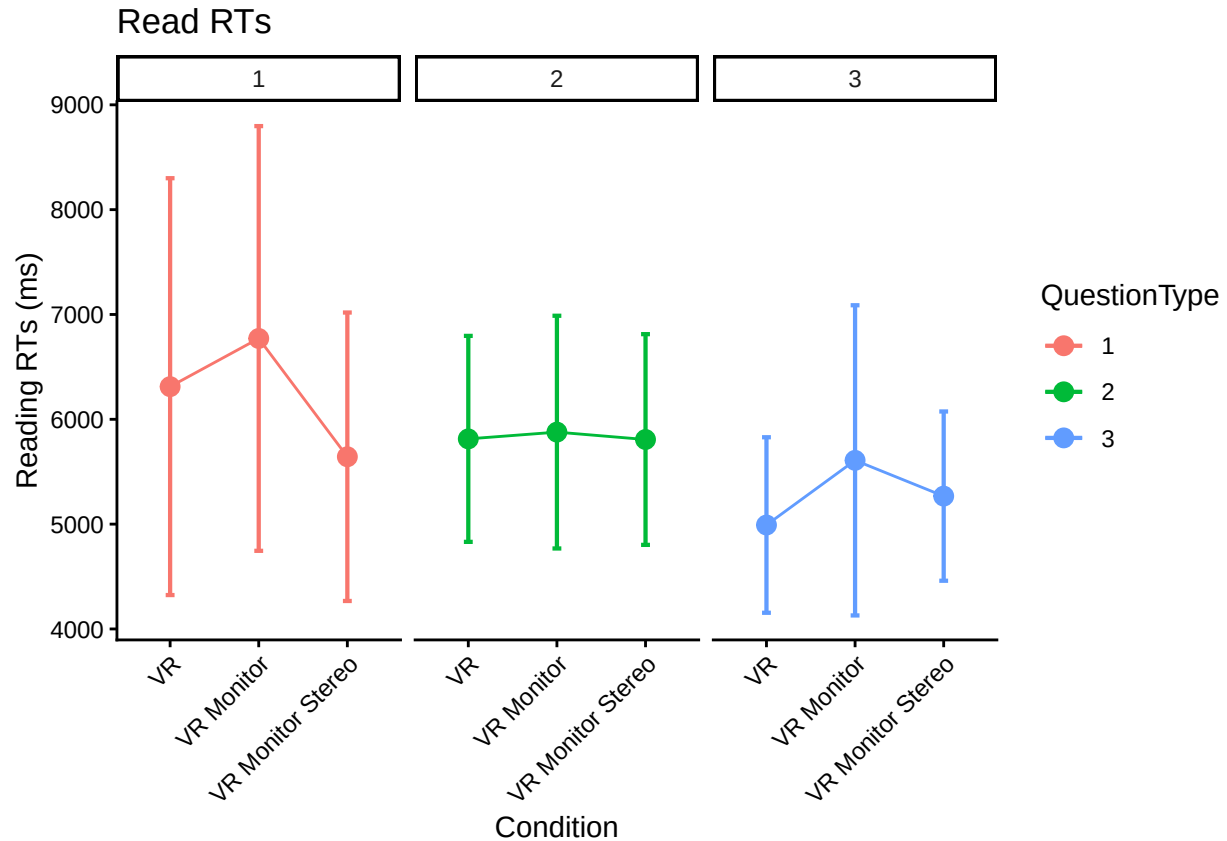


### Correct wider plot heterogeneous variance

The message from the superbPlot may indicate that some of the groups' variances are heterogeneous. We can try using the `tryon` purpose option as suggested, read Unequal variances, Welch test, Tryon adjustment, and superb.

```
# make the wide plot, use tryon for purpose as suggested because of group's heterogeneous variances
superbPlot(read_rt,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
  gamma = 0.95,
  adjustments = list(
    purpose = "tryon"
  ),
  plotLayout = "line",
  factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1)) +
facet_wrap(vars(QuestionType)) +
labs(y = "Reading RTs (ms)", title = "Read RTs")
```

## superb::FYI: The tryon adjustments per measures are Measure 1: 1.4938, Measure 2: 1.4938, Measure 3



### Perform anova and post-hoc tests

Finally perform anova on the read RT data and various posthoc tests.

```
# need back into long format for following, but we will use out dataframe
# will filled in na/missing and make wide again here
read_rt_long <- read_rt %>%
  pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_readRT")

read_rt_anova <- aov_ez(id = "ParticipantId",
  dv = "mean_readRT",
  data = read_rt_long,
  within = "TrialType",
  between = "Condition",
  anova_table = list(es = "pes") #might want to double-check these
)
```

## Converting to factor: Condition

## Contrasts set to contr.sum for the following variables: Condition

```
kable(nice(read_rt_anova))
```

| Effect              | df           | MSE         | F      | pes  | p.value |
|---------------------|--------------|-------------|--------|------|---------|
| Condition           | 2, 119       | 14643620.63 | 0.55   | .009 | .581    |
| TrialType           | 1.58, 188.04 | 7498202.73  | 4.57 * | .037 | .018    |
| Condition:TrialType | 3.16, 188.04 | 7498202.73  | 0.65   | .011 | .594    |

```

#posthoc test for trial type
pairs(emmeans(read_rt_anova, "TrialType"), adjust = "Tukey")

## contrast      estimate SE df t.ratio p.value
## Type1 - Type2      409 330 119   1.238  0.4333
## Type1 - Type3      953 373 119   2.553  0.0318
## Type2 - Type3      544 227 119   2.397  0.0472
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates

#posthoc test for condition, still using Tukey
pairs(emmeans(read_rt_anova, "Condition"), adjust = "Tukey")

## contrast      estimate SE df t.ratio p.value
## VR - VR Monitor      -380 472 119  -0.806  0.7002
## VR - VR Monitor Stereo    133 496 119   0.269  0.9610
## VR Monitor - VR Monitor Stereo    513 523 119   0.982  0.5894
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates

#posthoc test for condition, others seem to be scheffe, sidak, bonferroni, dunnett, mvt, none
pairs(emmeans(read_rt_anova, "Condition"), adjust = "scheffe")

## contrast      estimate SE df t.ratio p.value
## VR - VR Monitor      -380 472 119  -0.806  0.7234
## VR - VR Monitor Stereo    133 496 119   0.269  0.9646
## VR Monitor - VR Monitor Stereo    513 523 119   0.982  0.6184
##
## Results are averaged over the levels of: TrialType
## P value adjustment: scheffe method with rank 2

rm(read_rt_long)
rm(read_rt_anova)

```

## Alternative Analysis using glmm TMB package

The following analysis is for the heterogenous group condition analysis. See .

The following determines if a more complex model is justified.

```

model_glmm_hom = glmmTMB(mean_readRT ~ Condition * TrialType + (1 | ParticipantId),
  data=trial_type_means, family = Gamma(link="log"),
  dispformula = ~ 1, REML = FALSE)
model_glmm_hetero = glmmTMB(mean_readRT ~ Condition * TrialType + (1 | ParticipantId),
  data=trial_type_means, family = Gamma(link="log"),
  dispformula = ~ Condition, REML = FALSE)

anova(model_glmm_hom, model_glmm_hetero)

```

```

## Data: trial_type_means
## Models:
## model_glmm_hom: mean_readRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~1
## model_glmm_hetero: mean_readRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~Condition
##           Df    AIC    BIC logLik deviance Chisq Chi Df Pr(>Chisq)

```

```
## model_glmm_hom      11 6566.0 6608.8 -3272.0    6544.0
## model_glmm_hetero 13 6564.5 6615.0 -3269.2    6538.5 5.5347      2    0.06283 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### Try overall effect reporting

```
fixed_effects_anova <- Anova(model_glmm_hom, type="III")
kable(fixed_effects_anova)
```

|                     | Chisq        | Df | Pr(>Chisq) |
|---------------------|--------------|----|------------|
| (Intercept)         | 1.589899e+04 | 1  | 0.0000000  |
| Condition           | 7.826198e-01 | 2  | 0.6761706  |
| TrialType           | 6.785154e+00 | 2  | 0.0336219  |
| Condition:TrialType | 2.856415e+00 | 4  | 0.5821343  |

### EMM on Condition and Interaction

If the glmm analysis is not significant, just use the homogeneous model. Otherwise report on the more complex heterogeneous model?

```
emm_main_condition <- emmeans(model_glmm_hom,
                               specs = ~ Condition,
                               type = "response")
```

## NOTE: Results may be misleading due to involvement in interactions

```
condition_results <- pairs(emm_main_condition)
```

```
kable(condition_results)
```

| contrast                       | ratio     | SE        | df  | null | z.ratio    | p.value   |
|--------------------------------|-----------|-----------|-----|------|------------|-----------|
| VR / VR Monitor                | 0.9661949 | 0.0845442 | Inf | 1    | -0.3930159 | 0.9183917 |
| VR / VR Monitor Stereo         | 0.9801000 | 0.0900416 | Inf | 1    | -0.2187952 | 0.9739550 |
| VR Monitor / VR Monitor Stereo | 1.0143917 | 0.0981265 | Inf | 1    | 0.1477147  | 0.9880428 |

```
emm_interaction <- emmeans(model_glmm_hom,
                            specs = ~ Condition * TrialType,
                            type = "response")
```

```
simple_effects_results <- pairs(emm_interaction, by = "TrialType")
```

```
kable(simple_effects_results)
```

| contrast                       | TrialType | ratio     | SE        | df  | null | z.ratio    | p.value   |
|--------------------------------|-----------|-----------|-----------|-----|------|------------|-----------|
| VR / VR Monitor                | Type1     | 0.9428036 | 0.0966719 | Inf | 1    | -0.5744022 | 0.8337864 |
| VR / VR Monitor Stereo         | Type1     | 1.0400752 | 0.1118570 | Inf | 1    | 0.3653561  | 0.9290679 |
| VR Monitor / VR Monitor Stereo | Type1     | 1.1031726 | 0.1244085 | Inf | 1    | 0.8706858  | 0.6588260 |

Stereo

| contrast                       | TrialType | ratio     | SE        | df  | null | z.ratio    | p.value   |
|--------------------------------|-----------|-----------|-----------|-----|------|------------|-----------|
| VR / VR Monitor                | Type2     | 1.0153022 | 0.1038092 | Inf | 1    | 0.1485288  | 0.9879115 |
| VR / VR Monitor Stereo         | Type2     | 0.9675595 | 0.1042611 | Inf | 1    | -0.3060438 | 0.9496813 |
| VR Monitor / VR Monitor Stereo | Type2     | 0.9529769 | 0.1077057 | Inf | 1    | -0.4261589 | 0.9047567 |
| VR / VR Monitor                | Type3     | 0.9422748 | 0.0959314 | Inf | 1    | -0.5840223 | 0.8286865 |
| VR / VR Monitor Stereo         | Type3     | 0.9355537 | 0.0998279 | Inf | 1    | -0.6243092 | 0.8067746 |
| VR Monitor / VR Monitor Stereo | Type3     | 0.9928672 | 0.1118465 | Inf | 1    | -0.0635453 | 0.9977762 |

## Compare Answer RT

### Extract wide dataframe

Do all of the previous, but for the answer reaction times for the selected question types.

We again create a dataframe in needed format of only the answer RT data, filling in na/missing values in the data.

```
#compare answer times

trial_type_means <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_answerRT = mean(AnswerRT), n = n())

## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.

# first, make .csv files in wide format to double check in statview
answer_rt <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
  pivot_wider(names_from = TrialType, values_from = mean_answerRT)

# fill in na values with mean for type1, type2 and type3 RT
# the following is supposed to work, but we get NaN from calculation of mean here, not sure why...
#read_rt_type_wider_clean <- read_rt_type_wider %>% mutate(across(Type1, ~replace_na(., mean(., na.rm=T))
answer_rt <- answer_rt %>% replace_na(list(Type1 = mean(answer_rt$Type1, na.rm=TRUE)))
answer_rt <- answer_rt %>% replace_na(list(Type2 = mean(answer_rt$Type2, na.rm=TRUE)))
answer_rt <- answer_rt %>% replace_na(list(Type3 = mean(answer_rt$Type3, na.rm=TRUE)))

answer_rt_summary <- answer_rt %>%
  group_by(Condition) %>%
  summarize(meanType1 = mean(Type1, na.rm = T),
            meanType2 = mean(Type2, na.rm = T),
            meanType3 = mean(Type3, na.rm=T))
kable(answer_rt_summary)
```

| Condition         | meanType1 | meanType2 | meanType3 |
|-------------------|-----------|-----------|-----------|
| VR                | 22614.11  | 15243.81  | 11429.11  |
| VR Monitor        | 29303.13  | 16692.65  | 13590.41  |
| VR Monitor Stereo | 29819.09  | 15174.25  | 11997.74  |

```
rm(answer_rt_summary)

#rm(trial_type_means)
```

## Basic plots of main effect and wider format

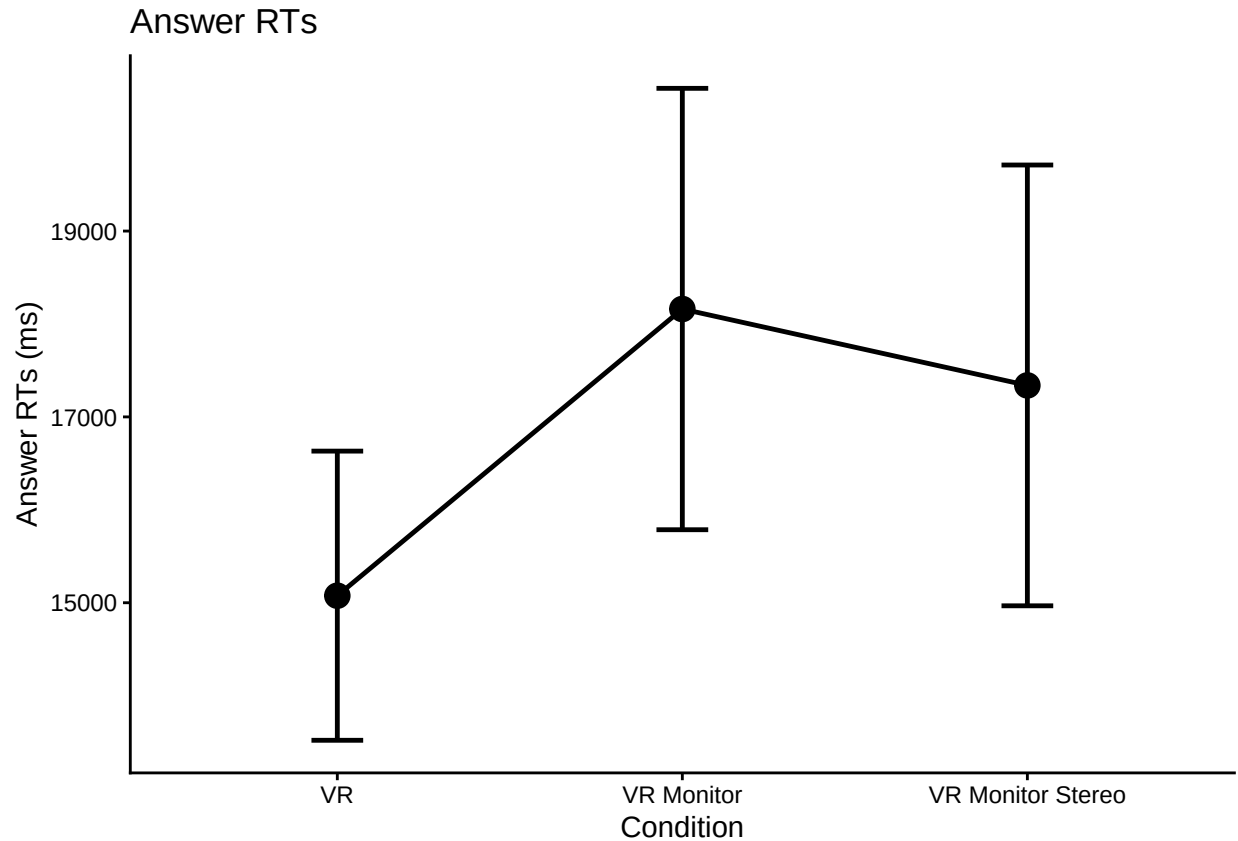
Plot of answer RT condition main effect, as well as wider format plot by the trial types.

```
#just condition main effect
answerplot1 <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition) %>%
  summarize(overallmean = mean(AnswerRT)) %>%
  group_by(Condition) %>%
  summarize(overall_condition_mean = mean(overallmean),
            se = std.error(overallmean),
            n = n(),
            CI = qt(0.975,df=n-1)*se)
```

## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups` argument.

```
ggplot(answerplot1, aes(Condition,
                        overall_condition_mean,
                        group = 1,
                        ymin = overall_condition_mean - CI,
                        ymax = overall_condition_mean + CI)) +

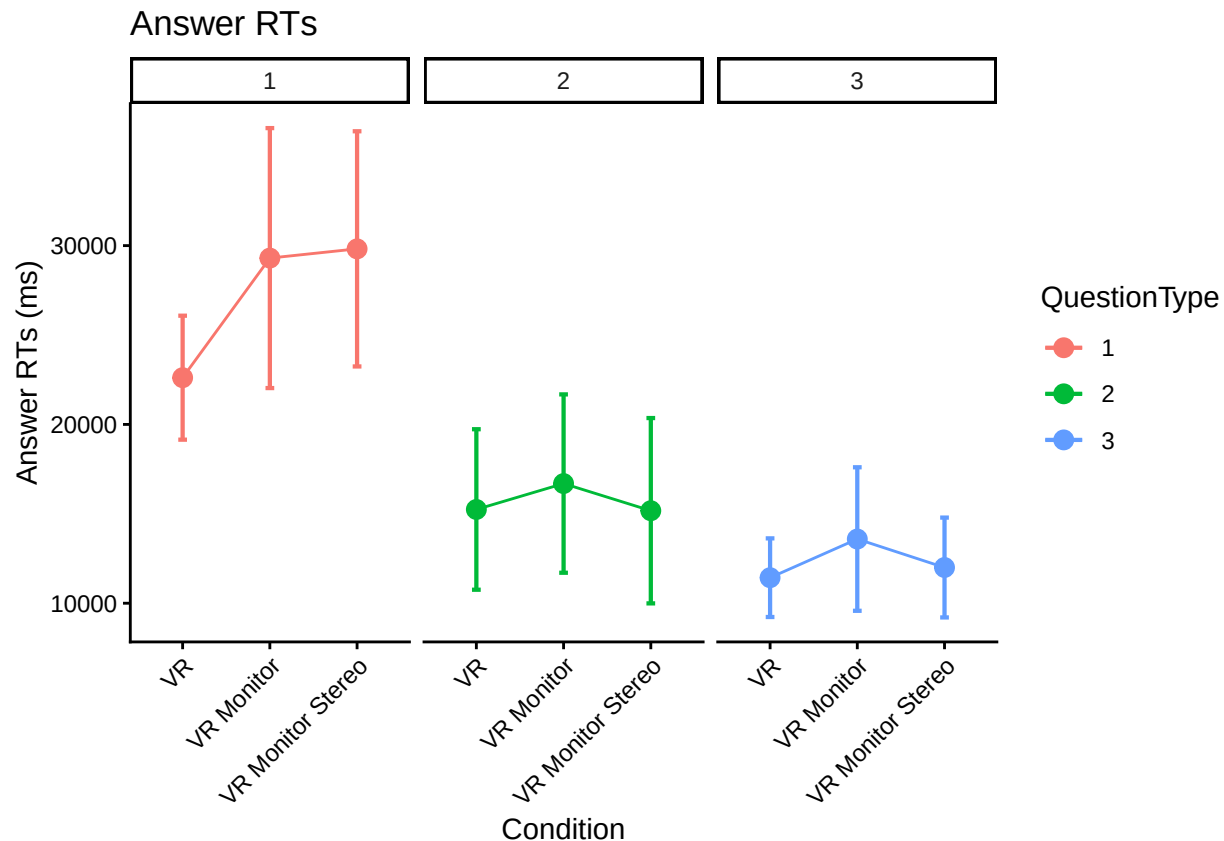
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Answer RTs (ms)", title = "Answer RTs")
```



```
rm(answerplot1)

#make a little plot using wide format
superbPlot(answer_rt,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
  gamma = 0.95,
  adjustments = list(
    purpose = "difference"
  ),
  plotLayout = "line",
  factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1)) +
facet_wrap(vars(QuestionType)) +
labs(y = "Answer RTs (ms)", title = "Answer RTs")
```

## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".



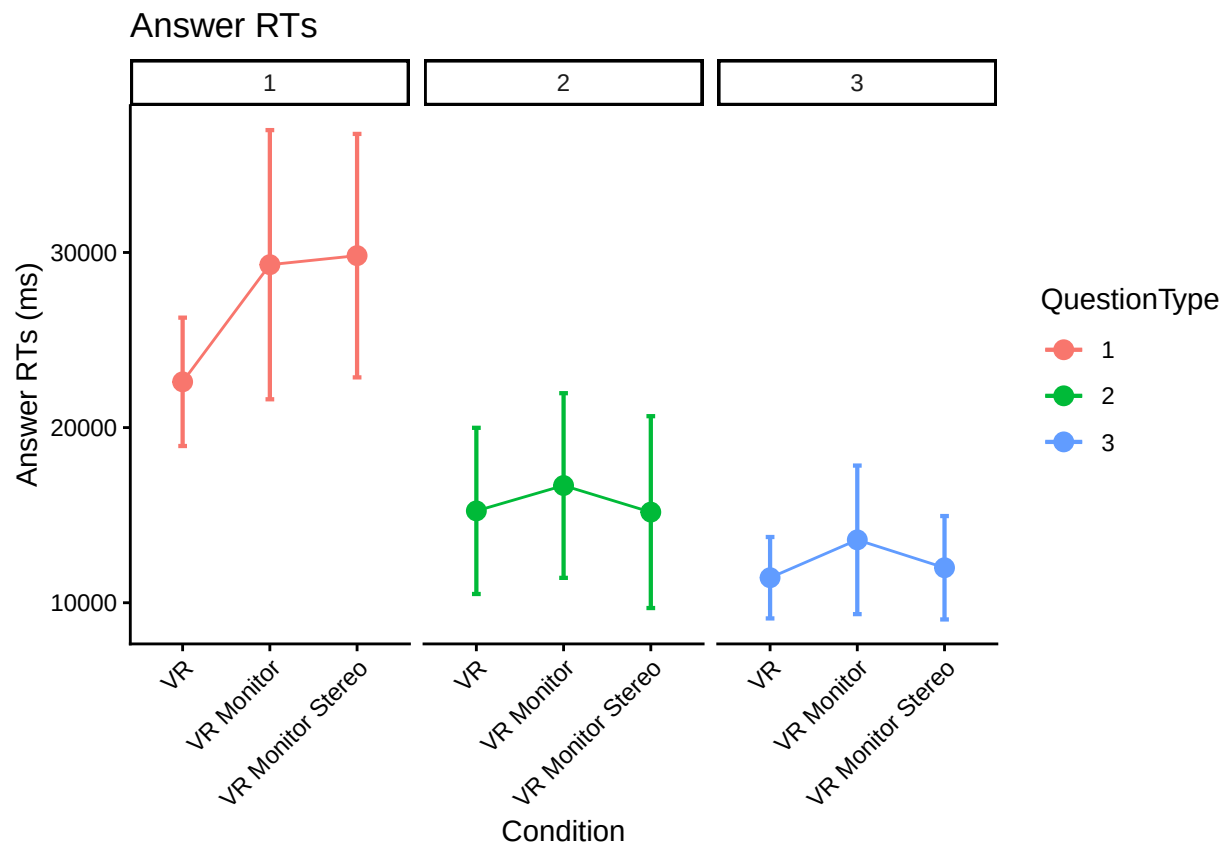
### Correct wider plot heterogeneous variance

The message from the superbPlot may indicate that some of the groups' variances are heterogeneous. We can try using the `tryon` purpose option as suggested, read Unequal variances, Welch test, Tryon adjustment, and superb.

```
# make the wide plot, use tryon for purpose as suggested because of group's heterogeneous variances
superbPlot(answer_rt,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
  gamma = 0.95,
  adjustments = list(
    purpose = "tryon"
  ),
  plotLayout = "line",
  factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1)) +
facet_wrap(vars(QuestionType)) +
labs(y = "Answer RTs (ms)", title = "Answer RTs")
```

## superb::FYI: The tryon adjustments per measures are Measure 1: 1.4950, Measure 2: 1.4950, Measure 3





### Perform anova and post-hoc tests

Finally perform anova on the answer RT data and various posthoc tests.

```
# need back into long format for following, but we will use out dataframe
# will filled in na/missing and make wide again here
answer_rt_long <- answer_rt %>%
  pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_answerRT")

answer_rt_anova <- aov_ez(id = "ParticipantId",
  dv = "mean_answerRT",
  data = answer_rt_long,
  within = "TrialType",
  between = "Condition",
  anova_table = list(es = "pes") #might want to double-check these
)
```

```
## Converting to factor: Condition
```

```
## Contrasts set to contr.sum for the following variables: Condition
```

```
kable(nice(answer_rt_anova))
```

| Effect              | df           | MSE          | F         | pes  | p.value |
|---------------------|--------------|--------------|-----------|------|---------|
| Condition           | 2, 119       | 148946021.28 | 2.87 +    | .046 | .060    |
| TrialType           | 1.76, 209.64 | 97205116.20  | 84.60 *** | .416 | <.001   |
| Condition:TrialType | 3.52, 209.64 | 97205116.20  | 2.12 +    | .034 | .088    |

```

#posthoc test for trial type
pairs(emmeans(answer_rt_anova, "TrialType"), adjust = "Tukey")

## contrast      estimate    SE  df t.ratio p.value
## Type1 - Type2    11542 1370 119   8.424 <.0001
## Type1 - Type3    14906 1220 119  12.184 <.0001
## Type2 - Type3     3364  980 119   3.432 0.0024
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates

#posthoc test for condition, still using Tukey
pairs(emmeans(answer_rt_anova, "Condition"), adjust = "Tukey")

## contrast      estimate    SE  df t.ratio p.value
## VR - VR Monitor    -3433 1510 119  -2.281 0.0625
## VR - VR Monitor Stereo -2568 1580 119  -1.625 0.2391
## VR Monitor - VR Monitor Stereo 865 1670 119   0.519 0.8622
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates

#posthoc test for condition, others seem to be scheffe, sidak, bonferroni, dunnett, mvt, none
pairs(emmeans(answer_rt_anova, "Condition"), adjust = "scheffe")

## contrast      estimate    SE  df t.ratio p.value
## VR - VR Monitor    -3433 1510 119  -2.281 0.0784
## VR - VR Monitor Stereo -2568 1580 119  -1.625 0.2709
## VR Monitor - VR Monitor Stereo 865 1670 119   0.519 0.8741
##
## Results are averaged over the levels of: TrialType
## P value adjustment: scheffe method with rank 2

rm(answer_rt_long)
rm(answer_rt_anova)

```

## Alternative Analysis using glmm TMB package

The following analysis is for the heterogenous group condition analysis. See .

The following determines if a more complex model is justified.

```

model_glmm_hom = glmmTMB(mean_answerRT ~ Condition * TrialType + (1 | ParticipantId),
  data=trial_type_means, family = Gamma(link="log"),
  dispformula = ~ 1, REML = FALSE)
model_glmm_hetero = glmmTMB(mean_answerRT ~ Condition * TrialType + (1 | ParticipantId),
  data=trial_type_means, family = Gamma(link="log"),
  dispformula = ~ Condition, REML = FALSE)

anova(model_glmm_hom, model_glmm_hetero)

## Data: trial_type_means
## Models:
## model_glmm_hom: mean_answerRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~1
## model_glmm_hetero: mean_answerRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~Condition
##              Df      AIC      BIC logLik deviance Chisq Chi Df Pr(>Chisq)

```

```
## model_glmm_hom      11 7517.9 7560.6 -3747.9 7495.9
## model_glmm_hetero 13 7521.4 7571.9 -3747.7 7495.4 0.4736      2      0.7892
```

### Try overall effect reporting

```
fixed_effects_anova <- Anova(model_glmm_hom, type="III")
kable(fixed_effects_anova)
```

|                     | Chisq        | Df | Pr(>Chisq) |
|---------------------|--------------|----|------------|
| (Intercept)         | 16539.823090 | 1  | 0.0000000  |
| Condition           | 6.164242     | 2  | 0.0458619  |
| TrialType           | 61.158375    | 2  | 0.0000000  |
| Condition:TrialType | 4.027610     | 4  | 0.4022821  |

### EMM on Condition and Interaction

If the glmm analysis is not significant, just use the homogeneous model. Otherwise report on the more complex heterogeneous model?

```
emm_main_condition <- emmeans(model_glmm_hom,
                               specs = ~ Condition,
                               type = "response")
```

## NOTE: Results may be misleading due to involvement in interactions

```
condition_results <- pairs(emm_main_condition)
```

```
summary(condition_results, level=0.95)
```

```
## contrast          ratio      SE df null z.ratio p.value
## VR / VR Monitor      0.857 0.0740 Inf    1  -1.782  0.1755
## VR / VR Monitor Stereo 0.902 0.0817 Inf    1  -1.137  0.4909
## VR Monitor / VR Monitor Stereo 1.052 0.1000 Inf    1   0.532  0.8555
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
```

```
#kable(condition_results)
```

```
emm_interaction <- emmeans(model_glmm_hom,
                           specs = ~ Condition * TrialType,
                           type = "response")
```

```
simple_effects_results <- pairs(emm_interaction, by = "TrialType")
```

```
summary(simple_effects_results, level=0.95)
```

```
## TrialType = Type1:
## contrast          ratio      SE df null z.ratio p.value
## VR / VR Monitor      0.789 0.0921 Inf    1  -2.026  0.1061
## VR / VR Monitor Stereo 0.768 0.0938 Inf    1  -2.161  0.0780
## VR Monitor / VR Monitor Stereo 0.973 0.1250 Inf    1  -0.216  0.9746
```

```
##
## TrialType = Type2:
## contrast ratio SE df null z.ratio p.value
## VR / VR Monitor 0.917 0.1080 Inf 1 -0.741 0.7390
## VR / VR Monitor Stereo 0.984 0.1230 Inf 1 -0.133 0.9902
## VR Monitor / VR Monitor Stereo 1.073 0.1390 Inf 1 0.542 0.8508
##
## TrialType = Type3:
## contrast ratio SE df null z.ratio p.value
## VR / VR Monitor 0.871 0.1010 Inf 1 -1.189 0.4600
## VR / VR Monitor Stereo 0.972 0.1180 Inf 1 -0.235 0.9700
## VR Monitor / VR Monitor Stereo 1.116 0.1430 Inf 1 0.853 0.6700
##
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
#kable(simple_effects_results)
```

## EMM

```
emm_main_condition <- emmeans(model_glmm_hom,
                              specs = ~ Condition,
                              type = "response")

## NOTE: Results may be misleading due to involvement in interactions
summary(emm_main_condition, level=0.95)

## Condition response SE df asymp.LCL asymp.UCL
## VR 15184 871 Inf 13568 16992
## VR Monitor 17708 1140 Inf 15602 20098
## VR Monitor Stereo 16832 1180 Inf 14667 19316
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
contrast_coefficients <- c(1, -0.5, -0.5)

# Create a named list for the contrast matrix
contrast_matrix <- list(
  VR_vs_Monitors_Collapsed = contrast_coefficients
)

comparison_results <- contrast(emm_main_condition,
                              method = contrast_matrix)
summary(comparison_results, level = 0.95)

## contrast ratio SE df null z.ratio p.value
## VR_vs_Monitors_Collapsed 0.879 0.0656 Inf 1 -1.723 0.0849
##
## Results are averaged over the levels of: TrialType
## Tests are performed on the log scale
contrast_coefficients <- c(-0.5, 1, -0.5)
```

```

# Create a named list for the contrast matrix
contrast_matrix <- list(
  VRMonitor_vs_Others_Collapsed = contrast_coefficients
)

comparison_results <- contrast(emm_main_condition,
                              method = contrast_matrix)
summary(comparison_results, level = 0.95)

## contrast          ratio      SE  df null z.ratio p.value
## VRMonitor_vs_Others_Collapsed  1.11 0.0873 Inf    1  1.297  0.1945
##
## Results are averaged over the levels of: TrialType
## Tests are performed on the log scale

```

## Compare Accuracy

### Extract wide dataframe

Do all of the previous again but on the Accuracy results.

We again create a dataframe in needed format of only the accuracy data.

A bit different from before, only need to create an average of the two correct features we have. But then we do a similar transformation into wide format and replace na/missing values.

```

all_data_no_outliers <- all_data_no_outliers %>%
  rowwise() %>%
  mutate(Correct_Avg = mean(c(Correct_1, Correct_2)))

trial_type_mean_accuracy <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_acc = mean(Correct_Avg),
            n = n())

```

## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using  
## the `.groups` argument.

```

accuracy <- trial_type_mean_accuracy %>%
  select(ParticipantId, Condition, TrialType, mean_acc) %>%
  pivot_wider(names_from = TrialType, values_from = mean_acc)

# replace na with mean
accuracy <- accuracy %>% replace_na(list(Type1 = mean(accuracy$Type1, na.rm=TRUE)))
accuracy <- accuracy %>% replace_na(list(Type2 = mean(accuracy$Type2, na.rm=TRUE)))
accuracy <- accuracy %>% replace_na(list(Type3 = mean(accuracy$Type3, na.rm=TRUE)))

rm(trial_type_mean_accuracy)

```

### Basic plots of main effect and wider format

```

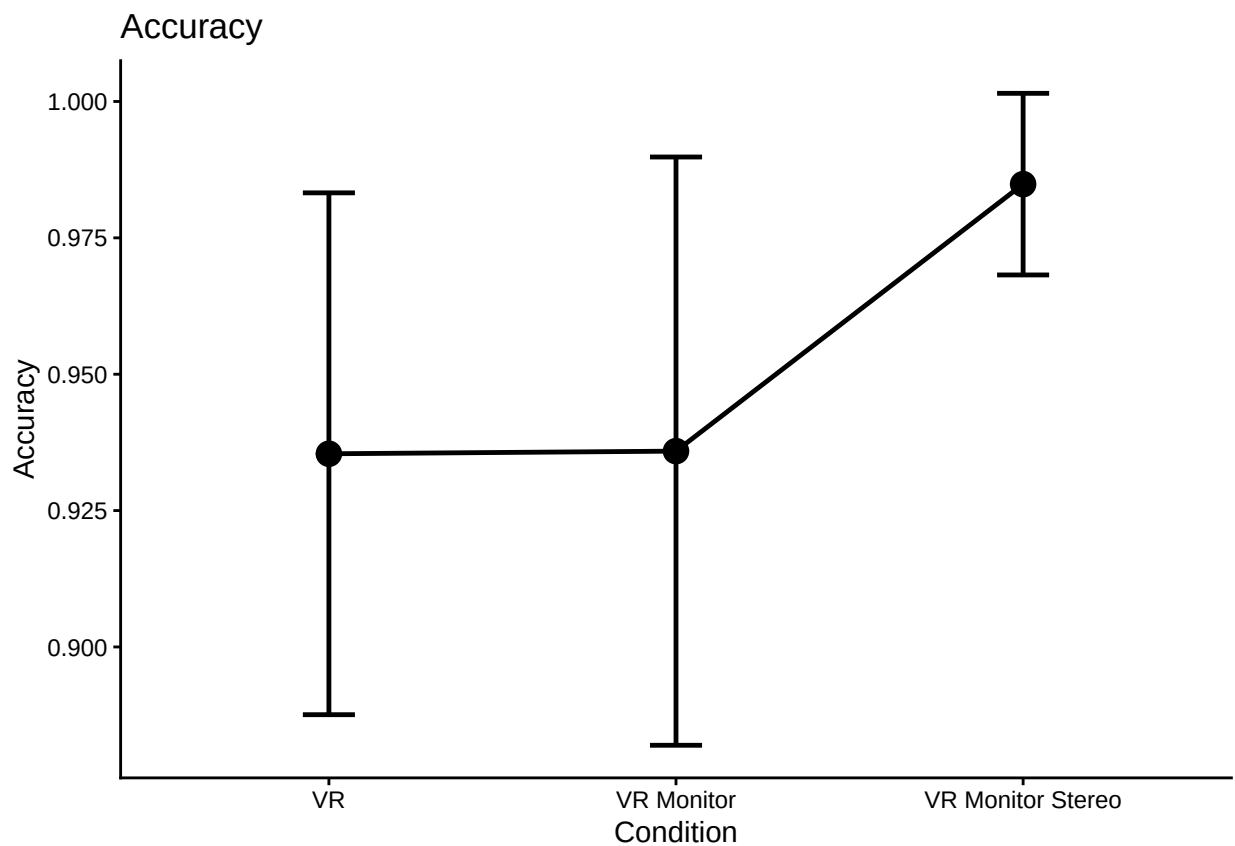
#just condition main effect
accplot1 <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition) %>%
  summarize(overallmean = mean(Correct_Avg)) %>%
  group_by(Condition) %>%
  summarize(overall_condition_mean = mean(overallmean),

```

```
se = std.error(overallmean),
n = n(),
CI = qt(0.975,df=n-1)*se)
```

## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups` argument.

```
ggplot(accplot1, aes(Condition,
                     overall_condition_mean,
                     group = 1,
                     ymin = overall_condition_mean - CI,
                     ymax = overall_condition_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Accuracy", title = "Accuracy")
```



```
rm(accplot1)

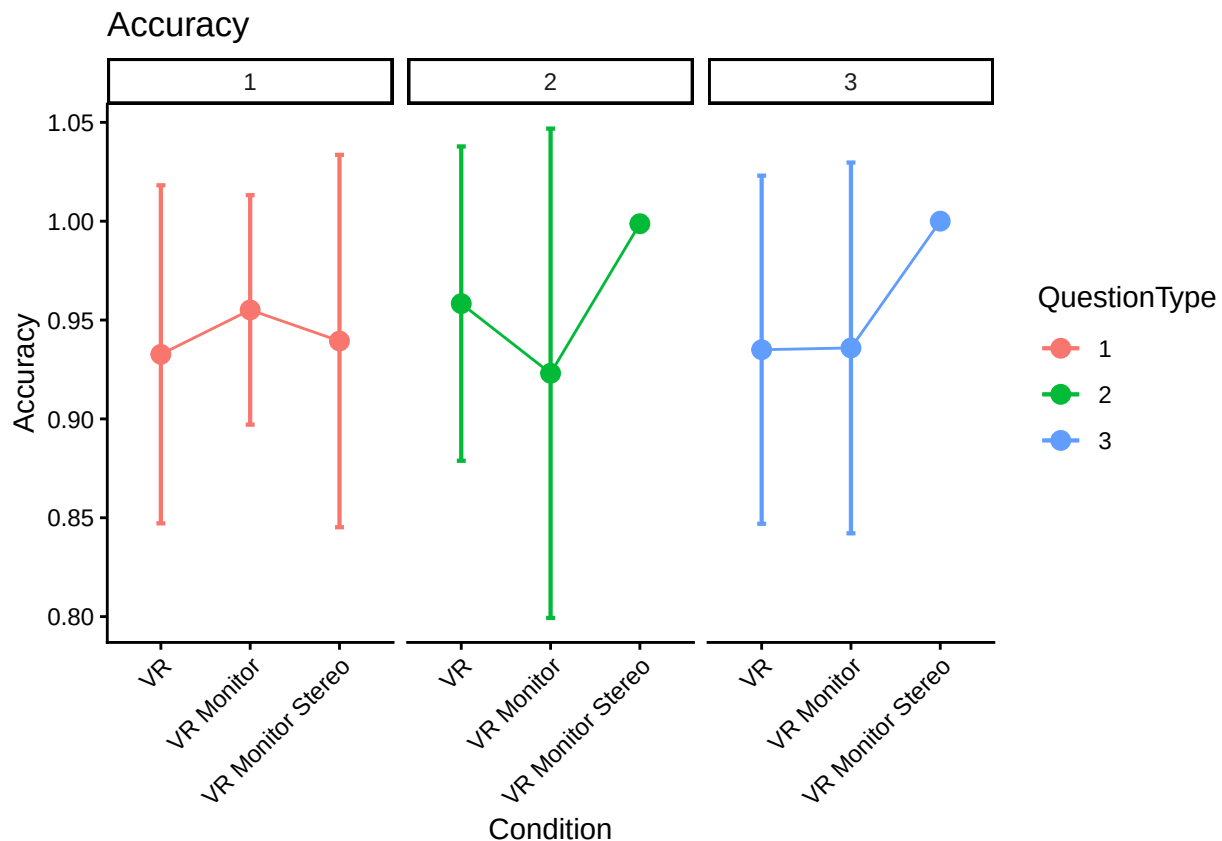
#make the little plot
superbPlot(accuracy,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
```

```

gamma      = 0.95,
adjustments = list(
  purpose   = "difference"
),
plotLayout = "line",
factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Accuracy", title = "Accuracy")

```

## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".



### Correct wider plot heterogeneous variance

The message from the superbPlot may indicate that some of the groups' variances are heterogeneous. We can try using the tryon purpose option as suggested, read Unequal variances, Welch test, Tryon adjustment, and superb.

```

# make the wide plot, use tryon for purpose as suggested because of group's heterogeneous variances
superbPlot(accuracy,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",

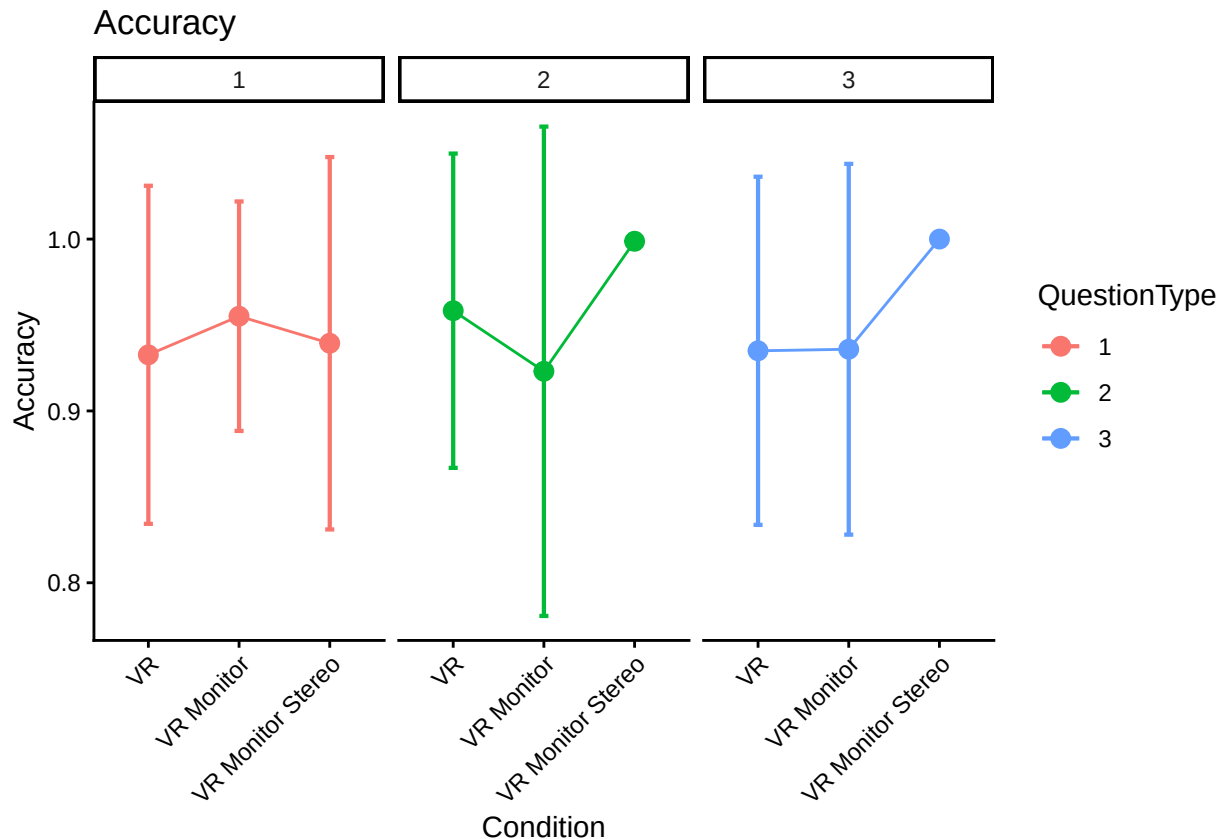
```

```

gamma      = 0.95,
adjustments = list(
  purpose   = "tryon"
),
plotLayout = "line",
factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Accuracy", title = "Accuracy")

```

## superb::FYI: The tryon adjustments per measures are Measure 1: 1.6270, Measure 2: 1.6270, Measure 3



### Perform anova and post-hoc tests

Finally perform anova on the accuracy data and various posthoc tests.

```

# need back into long format for following, but we will use out dataframe
# will filled in na/missing and make wide again here
accuracy_long <- accuracy %>%
  pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_acc")

accuracy_anova <- aov_ez(id = "ParticipantId",
  dv = "mean_acc",
  data = accuracy_long,
  within = "TrialType",

```



```

        between = "Condition",
        anova_table = list(es = "pes") #might want to double-check these
)

```

```
## Converting to factor: Condition
```

```
## Contrasts set to contr.sum for the following variables: Condition
```

```
kable(nice(accuracy_anova))
```

| Effect              | df           | MSE  | F    | pes  | p.value |
|---------------------|--------------|------|------|------|---------|
| Condition           | 2, 119       | 0.06 | 0.98 | .016 | .378    |
| TrialType           | 1.98, 235.25 | 0.02 | 0.43 | .004 | .648    |
| Condition:TrialType | 3.95, 235.25 | 0.02 | 1.05 | .017 | .382    |

```
#posthoc test for trial type
```

```
pairs(emmeans(accuracy_anova, "TrialType"), adjust = "Tukey")
```

```
## contrast      estimate      SE  df t.ratio p.value
## Type1 - Type2 -0.01764 0.0192 119  -0.919  0.6291
## Type1 - Type3 -0.01457 0.0207 119  -0.703  0.7622
## Type2 - Type3  0.00308 0.0210 119   0.147  0.9882
##
```

```
## Results are averaged over the levels of: Condition
```

```
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#posthoc test for condition, still using Tukey
```

```
pairs(emmeans(accuracy_anova, "Condition"), adjust = "Tukey")
```

```
## contrast              estimate      SE  df t.ratio p.value
## VR - VR Monitor          0.00396 0.0294 119   0.135  0.9901
## VR - VR Monitor Stereo  -0.03738 0.0309 119  -1.210  0.4493
## VR Monitor - VR Monitor Stereo -0.04134 0.0326 119  -1.269  0.4152
##
```

```
## Results are averaged over the levels of: TrialType
```

```
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#posthoc test for condition, others seem to be scheffe, sidak, bonferroni, dunnett, mvt, none
```

```
pairs(emmeans(accuracy_anova, "Condition"), adjust = "scheffe")
```

```
## contrast              estimate      SE  df t.ratio p.value
## VR - VR Monitor          0.00396 0.0294 119   0.135  0.9910
## VR - VR Monitor Stereo  -0.03738 0.0309 119  -1.210  0.4828
## VR Monitor - VR Monitor Stereo -0.04134 0.0326 119  -1.269  0.4492
##
```

```
## Results are averaged over the levels of: TrialType
```

```
## P value adjustment: scheffe method with rank 2
```

```
rm(accuracy_long)
```

```
rm(accuracy_anova)
```